

Long-term grazing effects on grassland soil properties in southern British Columbia

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Evans, C. R. W., Krzic, M., Broersma, K. and Thompson, D. J. 2012. **Long-term grazing effects on grassland soil properties in southern British Columbia.** *Can. J. Soil Sci.* **92**: 685–693. Although grazing effects on soil properties have been evaluated on various temperate grasslands, no study has dealt with these effects in the southern interior of British Columbia. The objective of this study was to determine the effects of spring versus fall season grazing as well as grazing [at a moderate rate of 0.6 animal unit months (AUM) ha⁻¹] versus non-grazing by beef cattle on selected soil properties. Effects were determined 20 and 30 yr after the establishment of the field experiment. Soil properties were determined for the 0- to 7.5-cm, 7.5- to 15-cm, and 15- to 30-cm depths. In comparison with fall grazing, spring grazing had greater soil bulk density, greater mechanical resistance within the top 15 cm of the soil profile, higher pH, and lower polysaccharides. This was true for both 20 and 30 yr of treatment. Grazing effects on aggregate stability were observed only after 30 yr with spring grazing leading to a more stable structure with a mean weight diameter (MWD) of 1.5 mm and 32% and 10% of aggregates in the 2- to 6-mm and 1- to 2-mm size fractions, respectively, compared with a MWD of 1.0 mm and 20% and 6% under fall grazing. Greater soil bulk density, mechanical resistance, and pH were observed under the grazed treatment relative to the control without grazing, but as we used a moderate stocking rate the impacts were not as great as in previous studies, which used heavy stocking rates. Our findings show that long-term grazing at a moderate stocking rate of 0.6 AUM ha⁻¹ did not have critical detrimental effects on soil properties as some land managers and ranchers have suggested.

Key words: Stocking rate, season of grazing, soil bulk density, aggregate stability, total soil carbon, root biomass

Evans, C. R. W., Krzic, M., Broersma, K. et Thompson, D. J. 2012. **Incidence de la paissance prolongée sur les propriétés du sol des prairies dans le sud de la Colombie-Britannique.** *Can. J. Soil Sci.* **92**: 685–693. Bien qu'on ait évalué l'incidence de la paissance sur les propriétés du sol des prairies dans divers climats tempérés, la question n'a pas vraiment été examinée dans l'intérieur méridional de la Colombie-Britannique. La présente étude devait préciser les conséquences d'une mise à l'herbe printanière plutôt qu'automnale ainsi que comparer l'incidence de la paissance par des bovins de boucherie (chargement modéré de 0,6 UAM par hectare) à l'absence de paissance sur certaines propriétés du sol. Les effets en question ont été établis 20 et 30 ans après le début de l'expérience sur le terrain. Les propriétés du sol ont été mesurées pour les couches de 0 à 7,5, de 7,5 à 15 et de 15 à 30 cm de profondeur. Comparativement à la paissance automnale, la mise à l'herbe printanière engendre une plus grande masse volumique apparente du sol, une résistance mécanique accrue de la couche de 15 cm supérieure du profil, une hausse du pH et une diminution de la concentration de polysaccharides. Ces observations sont valables pour les deux traitements de 20 et de 30 ans. L'incidence de la paissance sur la stabilité des agrégats n'apparaît qu'après 30 années, la mise à l'herbe au printemps donnant lieu à une structure plus stable, caractérisée par un diamètre pondéré moyen (DPM) de 1,5 mm, 32 % et 10 % des agrégats se retrouvant respectivement dans les fractions de 2–6 mm et de 1–2 mm, contre un DPM de 1,0 mm et une répartition de 20 % et de 6 % pour la mise à l'herbe à l'automne. La paissance donne lieu à une plus grande masse volumique apparente, à une résistance mécanique supérieure et à un pH plus élevé que dans le sol témoin, qui n'avait pas été pâturé. Néanmoins, les incidences sont plus faibles que celles rapportées dans des études antérieures, pour lesquelles la charge de bétail était plus grande. Ces constatations indiquent que la paissance à long terme avec un chargement modéré de 0,6 UAM par hectare ne nuit pas de manière dramatique aux propriétés du sol ainsi que certains exploitants de terres et grands éleveurs semblaient le craindre.

Mots clés: Chargement, saison de paissance, masse volumique apparente du sol, stabilité des agrégats, carbone total du sol, biomasse des racines

Grasslands occupy a large part of central North America, but those found in British Columbia are unique in a number of respects. These bunchgrass-dominated grasslands developed under the low-intensity grazing of

small populations of elk, deer, and bighorn sheep, as opposed to the prairie grasslands, which developed under the influence of large bison herds and periodic severe trampling. Consequently, bunchgrass species are generally less resilient to grazing pressures than prairie

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Abbreviations: AUM, animal unit months; MAP, mean annual precipitation; MWD, mean weight diameter

grass species (Mack and Thompson 1982). British Columbia's grasslands, which contain about 20% of wildlife species at risk (BC Ministry of Environment, Lands and Parks 1998), are facing a number of challenges to their long-term sustainability that includes overgrazing and more recently urban development, recreational activities, and fragmentation pressure. Stocking rate recommendations in British Columbia have historically relied on indicators of plant health, while soil properties have been mostly overlooked. In addition, the effects of grazing on soil properties of British Columbia's grasslands have not been well documented.

Generally, it has been reported that the trampling action of grazing animals impacts soil by increasing bulk density and mechanical resistance (Greenwood and McKenzie 2001) and reducing porosity, water infiltration (Warren et al. 1986), and aggregate stability (Warren et al. 1986; Broersma et al. 2000). Soil compaction of rangelands is often negatively correlated with soil organic matter content (Naeth et al. 1991a) and root growth (Greenwood and McKenzie 2001).

A substantial number of studies have evaluated the effects of grazing season and stocking rate on soil properties at semi-arid to sub-humid grassland sites, but these are largely from outside British Columbia. In a global review of rangeland literature comprising 236 sites, Milchunas and Lauenroth (1993) reported that responses of soil organic matter and total nitrogen to grazing rates were nearly equally divided between negative and positive, depending on the stocking rate, length of grazing experiment, climatic region, and plant species. For example, Schuman et al. (1999) found that total carbon in the top 30 cm soil was 21 and 22% higher after 12 yr of season-long cattle grazing at 0.67 and 2 AUM ha⁻¹, respectively, relative to non-grazed pasture on a Dark Brown Chernozem in Wyoming. Frank et al. (1995) reported a 25% decrease in total soil nitrogen over a period of 75 yr under a heavy (1.1 AUM ha⁻¹) cattle grazing treatment relative to a non-grazed control on a Dark Brown Chernozem in North Dakota. Some other soil properties have also been shown to be affected by grazing rates. For example, available phosphorus was 25% higher and mean weight diameter (MWD) was 15% lower in the Ah horizon under a heavy (2.4 AUM ha⁻¹) grazing treatment relative to a non-grazed control on a Black Chernozem in Alberta (Dormaar and Willms 1998). A light grazing rate of 1.2 AUM ha⁻¹ did not lead to a change in MWD relative to the control in a study by Lorenz and Rogler (1967) carried out on a fine silt Dark Brown Chernozem in North Dakota.

In the southern interior of British Columbia, there are concerns that cattle grazing is negatively impacting soil and water quality, ultimately leading to a decline in forage production for wildlife and domestic grazers (Wikeem and Wikeem 2004). No study, however, has evaluated the long-term impacts of grazing on soil properties within these unique ecosystems. This

first-of-its-kind study evaluated the effects of spring versus fall season grazing as well as grazing versus non-grazing by beef cattle (at a moderate rate of 0.6 AUM ha⁻¹) on selected grassland soil properties after 20 and 30 yr of treatment.

MATERIALS AND METHODS

Study Site

The study sites were located on the Lac du Bois Range (lat. 50°45'N, long. 120°25'W) north of Kamloops, BC, in Lac Du Bois Grasslands Provincial Park. This study was carried out within a broader long-term experiment established in 1978 and maintained by Agriculture and Agri-Food Canada's Kamloops Research Branch investigating the impact of cattle grazing on plant species composition. The study sites were located along an elevation gradient from 600 to 850 m above sea level on Brown and Dark Brown Chernozems. These soils developed from ablation till deposits associated with basic volcanic and limestone bedrock (Young et al. 1992) and are characterized by a loam texture (45% sand, 45% silt, and 10% clay) and an average coarse fragment (diameter > 2 mm) content of 7, 14, and 18% (volume) for the 0- to 7.5-cm, 7.5- to 15-cm, and 15- to 30-cm depths, respectively.

The study area is characterized by a semi-arid climate with a mean annual precipitation (MAP) of 270 mm, 32% of which falls between May and July. The area experiences hot, dry summers and moderately cold winters, often with very limited snowfall. Average annual air temperature in this area is 8.9°C, with an average July temperature of 21°C. The study sites are located within the Bunchgrass biogeoclimatic zone (Wikeem et al. 1993), where the dominant vegetation is the association of bluebunch wheatgrass [*Pseudoroegneria spicata* (Pursh) A. Löve syn. *Agropyron spicatum*] and basin big sagebrush (*Artemisia tridentata* Nutt. ssp. *tridentata*).

Experimental Design

Two pastures, each about 65 ha in size and each containing two 25 m × 50 m exclosures (fenced with 1.5-m-high barbed wire to exclude livestock grazing), were assigned either a spring or fall grazing treatment. This design was then repeated at two different locations within Lac Du Bois Grasslands Provincial Park. Grazing was performed at 0.6 AUM ha⁻¹ either in the spring (mid-April to mid-May) or the fall (mid-September to mid-October). The 0.6 AUM ha⁻¹ grazing rate (corresponding to 45–50% use of available forage) is considered to be a moderate rate for very dry bunchgrass grasslands in good condition (Holechek et al. 1999), as was the case on our study sites. Two 10-m-long transects were laid out, parallel to each other, within each non-grazed exclosure and directly outside of the exclosure within the grazed pastures. Transects were always laid out at least 3 m from the pasture or exclosure edge to avoid fence-line effects.

Soil sampling was done in early June 1998 and again in early June 2008. In both years, soil samples were collected at two random points along each of the transects. Thus, the number of samples (per depth of sampling) collected at each experimental unit was eight (2 locations $\times$ 2 transects $\times$ 2 samples/transect). Soil samples were collected from the 0- to 7.5-cm, 7.5- to 15-cm, and 15- to 30-cm depths, with the exception of those for aggregate stability analysis, which were collected only from the 0- to 7.5-cm depth. Identical laboratory procedures were used in both years except where noted.

Laboratory Analysis

Bulk density was determined on intact soil cores collected using a double-cylinder, drop-hammer sampler and 7.5-cm-diameter by 7.5-cm-deep cylinders. Samples were oven-dried at 105°C for 24 h and coarse fragments (diameter > 2 mm) within the sample were removed by sieve and weighed. The volume of coarse mineral fragments was determined from their dry mass based on a particle density of 2.65 g cm⁻³. Bulk density was calculated on a coarse fragment-free basis as the mass of dry soil per volume of field-moist soil (Blake and Hartge 1986).

Root biomass was determined on the same samples used for the bulk density determination. In 1998, samples were dispersed using calcium chloride, the dispersed soil was wet-sieved through a 1-mm sieve, and the roots remaining on the sieve were dried at 65°C for 48 h. Ash content was determined by ignition at 550°C for 4 h and root biomass is reported on an ash-free, oven-dry basis. In 2008, root biomass was determined using a method for rapid sample processing developed by Metcalfe et al. (2007). The mass of roots extracted over four 10-min periods was graphed and a logarithmic regression equation was fitted to plot extracted root mass to the point where the predicted extracted mass over one 10-min period was equal to 1% of the predicted total extracted mass. The accuracy of the logarithmic equation based on four intervals was checked using a series of extractions consisting of 12 intervals (120 min) (data not shown). The ash content of the roots was determined by ignition at 380°C for 12 h and root biomass is reported on an ash-free, oven-dry basis.

Soil samples for chemical analyses were collected with a hand trowel from the small soil pits excavated for extraction of intact core samples. Samples for chemical analysis were air-dried and passed through a 2-mm sieve. Total carbon (Nelson and Sommers 1982) and nitrogen (McGill and Figueiredo 1993) were determined by dry combustion using an automated elemental analyzer (LECO CNS-2000, Leco Corp., St. Joseph, MI). Soil polysaccharides were determined using the phenol-sulfuric acid method as described by Lowe (1993). Sodium bicarbonate extraction (NaHCO₃) (Olsen et al. 1954) was used to determine plant-available phosphorus. Soil pH was determined on a 1:2 (vol/vol) soil to 0.01 M CaCl₂ slurry (Hendershot and Lalande 1993).

Soil mechanical resistance (Bradford 1986) was recorded at 1.5-cm intervals down to a depth of 30 cm using a hand-pushed 13-mm diameter 30° cone penetrometer with an attached data logger (Agridy Rimik PTY Ltd. Toowoomba, QLD, Australia). Measurements were made at five random locations along each transect for a total of 10 measurements per pasture.

Aggregate stability was assessed using a variation of the wet sieving method (Nimmo and Perkins 2002). Samples were collected with a hand trowel from the 0- to 7.5-cm depth. Any organic matter present on the surface was scraped away before sampling. Two composite samples, each consisting of five individual subsamples, were collected along each transect. Samples were stored at 4°C until being passed through a 6-mm sieve and collected on a 2-mm sieve. Immediately before wet sieving, a pre-sieved 2–5-mm sample (of about 15 g) was placed on top of three nested sieves with openings of 2, 1, and 0.25 mm and moistened in a humidifier for 30 min to minimize slaking. Wet sieving was performed by a motor-driven mechanical device with a vertical stroke of 2.5 cm and an oscillating motion through an angle of 30° for 10 min at a rate of 30 strokes per minute. After the sieves were removed from the water, the material retained on each sieve was oven-dried at 105°C for 24 h and weighed. A correction was made to account for non-aggregate particles by crushing all material and washing the finer material through the sieve. Non-aggregate particles retained were weighed and their mass was subtracted from the total size fraction mass to determine the true mass of aggregates. The mass for each size fraction was expressed as a percentage of the total non-aggregate particle-free sample mass. Aggregate stability was expressed as the MWD, which represents the summation across size fractions of the product of the mean diameter of each size fraction and the proportion of the sample weight occurring in that size fraction.

Statistical Analyses

The data for each sampling depth were analyzed separately using the MIXED procedure in the statistical software package SAS (SAS Institute, Inc. 1990) with season, grazing, and their interactions in the model as fixed effects, and location and the location $\times$ season $\times$ grazing $\times$ transect interaction as random effects. This random interaction was used as the error term to test all fixed effects. Soil samples collected at consecutive depths from the same pit are dependent on each other; hence, soil data were analyzed separately for each depth of sampling.

The size of this long-term experiment made it impossible to replicate the treatments in the traditional way. However, given the large size of the plots used for the treatments and the large variability in the soil, the variation among transects is appropriate to test for the fixed effects. This approach allows testing the treatment means as if the experiment had been replicated

in a more traditional way. The UNIVARIATE procedure in the SAS package was used to check the residuals for normality and for potential outliers before performing the final analysis.

RESULTS

Grazing Effects on Soil Chemical Properties and Root Biomass

Spring grazing resulted in soil pH 0.2 to 0.4 units higher than fall grazing (Table 1). The grazing rate of 0.6 AUM ha⁻¹ resulted in a higher soil pH for the 0- to 7.5-cm

and 7.5- to 15-cm depths relative to the non-grazed control when measurements were done after 20 yr of grazing (Table 1), while there were no differences in soil pH after 30 yr of grazing. Available phosphorus was not affected by grazing, with the exception of the 0- to 7.5-cm depth after 30 yr of grazing where the fall-grazed treatment had 1.9 mg kg⁻¹ greater available phosphorus (Table 1).

Both total soil carbon and total soil nitrogen were generally unchanged by grazing either 20 or 30 yr after the establishment of the experiment on these semi-arid

Table 1. Grazing effects on soil chemical properties and root biomass as determined 20 yr (i.e., 1998) and 30 yr (i.e., 2008) after establishment of the field experiment on rangeland north of Kamloops, British Columbia

Year	Depth (cm)	Grazing season		Grazing	
		Spring	Fall	Without	With
Total carbon (kg m ⁻²)					
20	0–7.5	1.90 (0.10) ^z	2.40 (0.16) ^{*+}	2.09 (0.18)	2.21 (0.11)
	7.5–15	1.67 (0.09)	1.56 (0.10)	1.56 (0.11)	1.67 (0.09)
	15–30	2.68 (0.17)	2.27 (0.18) [*]	2.42 (0.15)	2.54 (0.21)
30	0–7.5	1.86 (0.16)	2.14 (0.13)	2.07 (0.16)	1.92 (0.14)
	7.5–15	1.46 (0.07)	1.57 (0.08)	1.59 (0.08)	1.44 (0.07)
	15–30	2.66 (0.19)	2.44 (0.18)	2.71 (0.19)	2.39 (0.18)
Total nitrogen (kg m ⁻²)					
20	0–7.5	0.17 (0.01)	0.20 (0.01) ⁺	0.18 (0.01)	0.19 (0.01)
	7.5–15	0.15 (0.01)	0.14 (0.01)	0.14 (0.01)	0.15 (0.01)
	15–30	0.24 (0.01)	0.20 (0.01) ⁺	0.22 (0.01)	0.22 (0.01)
30	0–7.5	0.17 (0.01)	0.19 (0.01)	0.19 (0.01)	0.17 (0.01)
	7.5–15	0.14 (0.01)	0.15 (0.02)	0.15 (0.01)	0.14 (0.01)
	15–30	0.24 (0.01)	0.24 (0.02)	0.25 (0.02)	0.23 (0.02)
Soil polysaccharides (kg m ⁻²)					
20	0–7.5	0.46 (0.02)	0.54 (0.02) [*]	0.50 (0.02)	0.50 (0.03)
	7.5–15	0.39 (0.02)	0.39 (0.02)	0.39 (0.03)	0.39 (0.02)
	15–30	0.66 (0.03)	0.75 (0.06) ⁺	0.69 (0.05)	0.73 (0.08)
30	0–7.5	0.59 (0.07)	0.83 (0.05) [*]	0.79 (0.06)	0.63 (0.07) [†]
	7.5–15	0.47 (0.05)	0.70 (0.06) ^{**}	0.62 (0.06)	0.54 (0.06)
	15–30	1.09 (0.09)	1.07 (0.09)	1.19 (0.07)	0.97 (0.10)
Soil pH in CaCl ₂					
20	0–7.5	6.3 (0.06)	6.1 (0.06) ^{**}	6.1 (0.06)	6.3 (0.07) [*]
	7.5–15	6.3 (0.08)	6.2 (0.05)	6.1 (0.07)	6.3 (0.05) [*]
	15–30	6.4 (0.06)	6.3 (0.09)	6.3 (0.06)	6.4 (0.08)
30	0–7.5	6.6 (0.09)	6.3 (0.06) ^{**}	6.4 (0.10)	6.4 (0.07)
	7.5–15	6.7 (0.09)	6.3 (0.06) ^{**}	6.5 (0.09)	6.5 (0.09)
	15–30	6.7 (0.10)	6.3 (0.08) ^{**}	6.5 (0.18)	6.5 (0.09)
Available phosphorus (mg kg ⁻¹ soil)					
20	0–7.5	13.3 (1.18)	14.1 (1.36)	14.4 (1.33)	13.0 (1.20)
	7.5–15	7.4 (0.82)	8.9 (1.19)	8.0 (0.97)	8.4 (1.10)
	15–30	5.8 (0.86)	7.0 (0.75)	6.7 (0.84)	6.0 (0.76)
30	0–7.5	9.1 (0.98)	11.0 (0.72) ⁺	10.0 (0.92)	10.1 (0.87)
	7.5–15	6.2 (0.63)	6.5 (0.59)	6.4 (0.63)	6.3 (0.59)
	15–30	4.9 (0.74)	6.4 (0.51)	5.3 (0.54)	5.9 (0.77)
Root biomass (Mg ha ⁻¹)					
20	0–7.5	14.9 (1.98)	19.9 (6.25)	16.1 (2.1)	18.7 (3.27)
	7.5–15	6.2 (0.62)	5.6 (0.70)	5.7 (0.49)	6.1 (0.8)
	15–30	12.7 (1.53)	8.9 (1.00) [*]	10.8 (1.38)	10.8 (1.4)
30	0–7.5	10.9 (2.18)	11.8 (1.80)	9.8 (1.17)	12.9 (2.52)
	7.5–15	5.2 (0.59)	7.5 (1.20)	7.4 (1.23)	5.7 (0.58)
	15–30	8.4 (0.75)	10.5 (1.13) ⁺	10.4 (1.14)	8.5 (0.76)

[‡]Values in parentheses are the standard error of the mean ($n = 16$).

⁺, ^{*}, ^{**} Significant at $P < 0.10$, $P < 0.05$, and $P < 0.01$, respectively, between the paired treatments (i.e., spring grazing vs. fall grazing and without grazing vs. with grazing).

grasslands (Table 1). In addition to determining total soil carbon and nitrogen as indicators of the overall quantity of the soil organic matter, analyses for soil polysaccharides were also included to reflect the more labile pool of soil organic matter. Soil polysaccharides were generally greater on pastures grazed in the fall during the dormant part of the growing season than on spring-grazed pastures (Table 1). On the other hand, levels of polysaccharides were similar in pastures with and without grazing (Table 1).

Root biomass did not significantly differ between spring and fall grazing treatments (Table 1) or between pastures with and without grazing. The only exception was observed at the 15- to 30-cm depth after 20 yr of grazing where root biomass under spring grazing was 12.7 Mg ha^{-1} relative to 8.9 Mg ha^{-1} under fall grazing.

Grazing Effects on Soil Physical Properties

Soil bulk density was generally greater for spring- than fall-grazed pastures (Table 2). The differences between treatments with and without grazing were less pronounced, but significantly greater soil bulk density was observed in the top 0–7.5 cm on pastures with grazing (this was true after both 20 and 30 yr).

The season of grazing had a significant effect on soil mechanical resistance after 20 yr of grazing, when 11–32% greater mechanical resistance was observed

under the spring-grazed treatment in the top 15 cm of the soil profile (Fig. 1). After 30 yr of grazing, soil mechanical resistance of spring- and fall-grazed pastures was generally similar. Greater mechanical resistance was determined on grazed plots in the top 15–17 cm of the soil profile relative to the non-grazed control (Fig. 2). Under all treatments in this study, the commonly cited critical limit for restricted root growth of 2500 kPa (Busscher 1990; Carter 2002) was reached at about 8 cm. Reflecting this, a strong negative correlation ($r = -0.57$) between mechanical resistance and root biomass was also found (Table 3).

The exclusion of cattle grazing for either 20 or 30 yr did not lead to significant differences in MWD or for any of the four aggregate size fractions relative to pastures that were grazed at 0.6 AUM ha^{-1} (Table 2). Twenty years of spring and fall grazing did not affect MWD or 2- to 6-mm, 0.25- to 1-mm, and <0.25 -mm size fractions, and only the proportion of the total aggregate sample present in the 1- to 2-mm fraction showed a significant difference since it was greater on spring- than fall-grazed treatment (Table 2). However, after 30 yr of grazing, differences in aggregate stability between the two seasons of grazing emerged. Spring grazing had 50% greater MWD relative to fall-grazed pastures and the proportions of the two largest aggregate size fractions (2–5 and 1–2 mm) were also greater

Table 2. Grazing effects on soil bulk density, aggregate mean weight diameter (MWD), and fraction of total soil sample present in four aggregate size classes (2–6, 1–2, 0.25–1, and <0.25 mm) as determined 20 yr (i.e., 1998) and 30 yr (i.e., 2008) after establishment of the field experiment on rangeland north of Kamloops, British Columbia

		Grazing season		Grazing	
Year	Depth (cm)	Spring	Fall	Without	With
<i>Bulk density ($Mg\ m^{-3}$)</i>					
20	0–7.5	1.09 (0.02) ^z	1.03 (0.02) ^{*+}	1.03 (0.02)	1.08 (0.02) [*]
	7.5–15	1.07 (0.02)	1.02 (0.03) ⁺	1.03 (0.03)	1.06 (0.02)
	15–30	1.09 (0.02)	0.98 (0.03) ^{**}	1.02 (0.02)	1.05 (0.03)
30	0–7.5	0.94 (0.04)	0.90 (0.02)	0.87 (0.03)	0.98 (0.03) ^{**}
	7.5–15	0.91 (0.02)	0.90 (0.02)	0.89 (0.02)	0.92 (0.02)
	15–30	0.92 (0.02)	0.87 (0.03) ⁺	0.89 (0.02)	0.90 (0.02)
<i>MWD (mm)</i>					
20	0–7.5	2.3 (0.09)	2.3 (0.11)	2.4 (0.09)	2.2 (0.11)
30	0–7.5	1.5 (0.10)	1.0 (0.07) ^{**}	1.3 (0.11)	1.2 (0.11)
<i>2–6 mm size fraction ($kg\ kg^{-1}\ soil$)</i>					
20	0–7.5	0.52 (0.025)	0.53 (0.030)	0.54 (0.025)	0.50 (0.030)
30	0–7.5	0.32 (0.025)	0.20 (0.018) ^{**}	0.27 (0.026)	0.25 (0.027)
<i>1–2 mm size fraction ($kg\ kg^{-1}\ soil$)</i>					
20	0–7.5	0.10 (0.004)	0.09 (0.005) [*]	0.09 (0.005)	0.09 (0.004)
30	0–7.5	0.10 (0.007)	0.06 (0.004) ^{**}	0.08 (0.006)	0.08 (0.009)
<i>0.25–1 mm size fraction ($kg\ kg^{-1}\ soil$)</i>					
20	0–7.5	0.07 (0.005)	0.07 (0.004)	0.06 (0.004)	0.07 (0.005)
30	0–7.5	0.08 (0.005)	0.08 (0.003)	0.08 (0.004)	0.08 (0.005)
<i><0.25 mm size fraction ($kg\ kg^{-1}\ soil$)</i>					
20	0–7.5	0.32 (0.020)	0.32 (0.024)	0.30 (0.019)	0.34 (0.024)
30	0–7.5	0.50 (0.024)	0.66 (0.019) ^{**}	0.57 (0.030)	0.59 (0.030)

^zValues in parentheses are the standard error of the mean ($n = 16$).

⁺, ^{*}, ^{**} Significant at $P < 0.10$, $P < 0.05$, and $P < 0.01$, respectively, between the paired treatments (i.e., spring grazing vs. fall grazing and without grazing vs. with grazing).

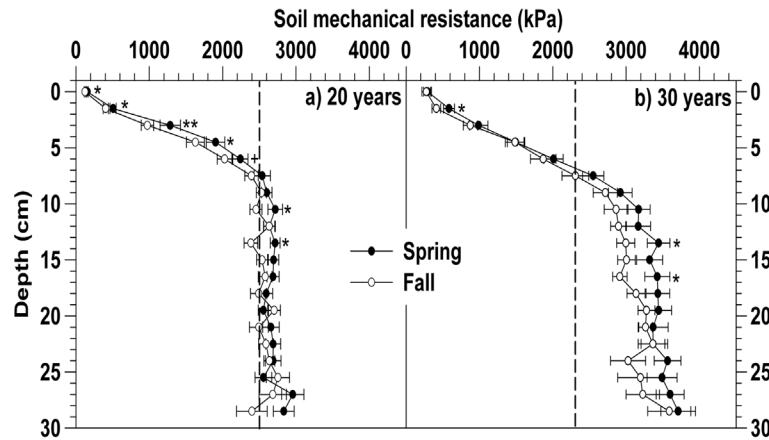


Fig. 1. Soil mechanical resistance under spring and fall grazing treatments as determined 20 yr (i.e., 1998) (a) and 30 yr (i.e., 2008) (b) after establishment of experiment on rangeland north of Kamloops, British Columbia. The dashed line represents the critical upper limit for root growth. Error bars represent the standard error of the mean ($n=16$). +, *, and ** indicate significance at $P<0.10$, $P<0.05$, and $P<0.01$, respectively.

under spring relative to fall grazing (Table 2). At the same time, the proportion of the total aggregate sample present as the smallest aggregate size (<0.25 mm) was lower in spring-grazed pastures (Table 2). These data suggest that grazing carried out in the spring favours the creation of larger aggregate size fractions on these semi-arid rangelands.

DISCUSSION

Grazing Effects on Chemical Properties and Root Biomass

Spring and fall grazing at a moderate rate of 0.6 AUM ha^{-1} for either 20 or 30 yr on these bunchgrass rangelands in the southern interior of British Columbia generally did not change the soil chemical properties that were measured in this study. The only soil chemical properties that were affected were pH (greater on the

spring-grazed pastures) and polysaccharides (greater on the fall-grazed pastures).

An increase in soil pH of the same magnitude as observed in our study was also reported in grasslands in Alberta by Dormaar and Willms (1998) and Donkor et al. (2002). In those studies, the removal of topsoil due to grazing-enhanced erosion and the lowering of the soil surface towards more alkaline subsoil contributed to the rise in soil pH in grazed plots. It is possible that something similar occurred on our study sites, since spring-grazed pastures were more compacted (as shown by greater bulk density and mechanical resistance) than those grazed later in the season.

An interesting finding of this study was that the elimination of grazing for either 20 or 30 yr on these semi-arid grasslands did not lead to an increase of soil carbon relative to the pastures grazed at a moderate rate.

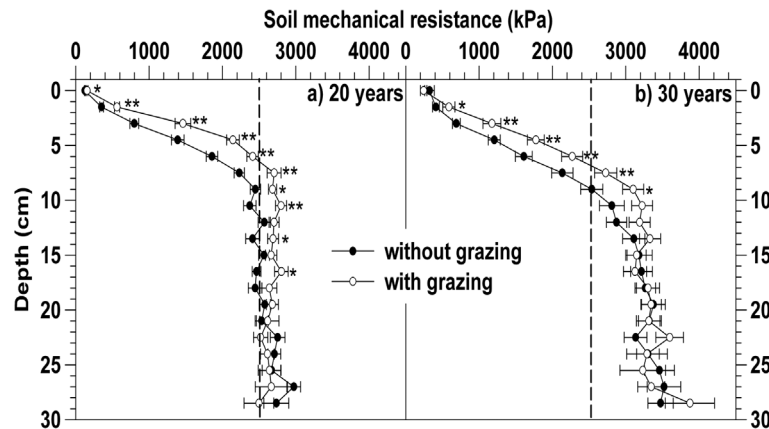


Fig. 2. Soil mechanical resistance under treatments with and without grazing as determined 20 yr (i.e., 1998) (a) and 30 yr (i.e., 2008) (b) after establishment of experiment on rangeland north of Kamloops, British Columbia. The dashed line represents the critical upper limit for root growth. Error bars represent the standard error of the mean ($n=16$). +, *, and ** indicate significance at $P<0.10$, $P<0.05$, and $P<0.01$, respectively.

Table 3. Correlation coefficients among aggregate stability parameters and selected soil properties. All data (from 1998 and 2008) are from the 0- to 7.5-cm depth and $n = 16$ for all correlations

Property	MWD		2- to 6-mm aggregates		1- to 2-mm aggregates		0.25- to 1-mm aggregates		<0.25-mm aggregates	
	<i>r</i>	<i>P</i>	<i>r</i>	<i>P</i>	<i>r</i>	<i>P</i>	<i>r</i>	<i>P</i>	<i>r</i>	<i>P</i>
Polysaccharides	-0.59	0.017	-0.56	0.023	-0.68	0.004	0.04	0.885	0.64	0.007
Total carbon	0.07	0.799	0.08	0.760	-0.22	0.420	-0.38	0.143	-0.02	0.927
Total nitrogen	-0.01	0.977	0.01	0.983	-0.23	0.398	-0.37	0.155	0.05	0.846
Root biomass	0.54	0.032	0.54	0.030	0.09	0.728	-0.42	0.105	-0.52	0.039
Bulk density	0.56	0.025	0.54	0.030	0.50	0.051	-0.09	0.747	-0.60	0.015
Mechanical resistance	0.02	0.934	0.00	0.990	0.41	0.116	0.19	0.480	-0.07	0.802

Some other grassland studies (Dormaer et al. 1977; Johnston et al. 1971) have also reported a lack of change in soil organic matter under light to moderate grazing relative to non-grazed controls. However, studies investigating lightly grazed treatments have more often reported an increase in soil organic matter (Manley et al. 1995; Ganjegunte et al. 2005). For example, a stocking rate of 0.67 AUM ha⁻¹ carried out for a whole season on a sandy loam Brown Chernozem in Wyoming led to 21% higher total soil carbon (and 29% higher total soil nitrogen) relative to a non-grazed control (Schuman et al. 1999). The authors attributed the increase in soil carbon and nitrogen under grazing to an increase in blue grama [*Bouteloua gracilis* (H.B.K.) Lag. Ex Steud.] cover, which allocates more carbon and nutrients to below-ground parts than other species characteristic of the mixed-grass prairie. Other factors, such as increased litter decomposition and litter incorporation due to cattle trampling, have also been mentioned.

Polysaccharides, which are long-chained carbohydrates, comprise on average about 10% of soil organic matter (Cheshire et al. 1973). Soil polysaccharides are mainly the result of two continuous processes: (1) the degradation of plant, animal, and microbial polysaccharides and (2) de novo synthesis by soil organisms using various organic compounds. It is possible that grazing during the dormant part of the season (i.e., fall), which allowed unobstructed plant growth earlier in the season, led to a greater amount of plant exudates released from roots and in turn enhanced microbial activity and polysaccharides synthesis.

Our finding that only soil polysaccharides, but not total soil carbon and nitrogen, were affected by grazing was in agreement with observations made by Carter (2002). Carter stated that it is generally more challenging to detect small changes in soil organic matter against a relatively large background mass and that attributes of soil organic matter (e.g., light organic matter fraction, polysaccharides, microbial biomass) are therefore more sensitive to effects of management practices. The lack of differences for total soil carbon and nitrogen is not surprising since the study sites were in good condition and the level of use had been fairly light with grazing occurring for only a month per growing season at a relatively moderate rate of 0.6 AUM ha⁻¹.

Differences in root biomass between different grazing treatments were not observed. The overall lack of differences for root biomass data is supported by the findings for total soil carbon and nitrogen, which also exhibited no significant differences between grazing treatments. This is not surprising since roots are the main source of soil carbon in grasslands. A lack of grazing impact on root biomass has also been reported by Schuman et al. (1999) on a Dark Brown Chernozem at a semi-arid site (384 mm MAP) in Wyoming after 12 yr of season-long cattle grazing at 0.6 AUM ha⁻¹ relative to a control. Studies that have reported changes in root biomass due to grazing have generally used heavy stocking rates, and those changes were often accompanied by changes in plant species composition from deep-rooted perennial species to shallow-rooted weedy species (Naeth et al. 1990; Snyman 2005). For example, heavy (5 AUM ha⁻¹) grazing resulted in an increase in root mass in the top 15 cm from 17.2 dry matter tons ha⁻¹ on pasture without grazing to 25.3 dry matter tons ha⁻¹ on heavily grazed pasture on a Black Chernozem (500 mm MAP) in Alberta (Dormaer et al. 1977). This increase was correlated to a change in dominant vegetation from rough fescue (*Festuca campestris* Rydb.) to forbs and shrubs. In our study, 40% of root biomass was found in the top 15 cm under both treatments with and without grazing (true for both years). This indicates that grazing most likely did not lead to a substantial shift in plant species composition from the deep-rooted bunchgrasses typical of these grasslands to more shallow-rooted species.

Effects of Grazing on Soil Physical Properties

Soil physical properties exhibited more pronounced changes under long-term grazing than soil chemical properties. Both soil bulk density and mechanical resistance were greater when grazing was done earlier in the growing season rather than later, which is in agreement with several other studies carried out in arid to semi-arid grassland areas. Studies by Laycock and Conrad (1967) and Naeth et al. (1991b) reported more severe impacts of early season grazing because soils were moister in the spring from snowmelt and immature vegetation was less resilient to grazing and trampling. Time of grazing at light (1.5 AUM ha⁻¹) and heavy (4.4 AUM ha⁻¹)

stocking rates was the focus of a study carried out on a loam Black Chernozem at a foothills fescue site in northeast Alberta (380 mm MAP) by Naeth et al. (1990). They found that bulk density in the top 7.5 cm was 8% greater after 12 yr of heavy spring grazing relative to heavy fall grazing. However, light grazing done either in the spring or fall did not affect soil bulk density of the 0- to 7.5-cm depth. Numerous studies (Dormaar et al. 1989; Naeth et al. 1990; Mapfumo et al. 1999) have reported greater bulk density and mechanical resistance on pastures with grazing relative to controls without grazing. Evidence for this tendency is reinforced by our finding that long-term grazing at the moderate rate used in our study resulted in greater soil mechanical resistance (in top 15 cm) and bulk density (at the 0- to 7.5-cm depth).

Contrary to the greater soil compaction under the spring grazing treatment as indicated by higher bulk density and mechanical resistance, aggregate stability data show a different trend. After 20 yr of grazing, aggregate stability was similar under all grazing treatments. Differences were observed 10 yr later, when spring grazing resulted in greater aggregate stability (i.e., greater MWD, a greater proportion of 2- to 6-mm and 1- to 2-mm aggregates and a lower proportion of the <0.25-mm size fraction) relative to fall grazing. Aggregate MWD has generally been shown to decrease under grazing pressure (Dormaar and Willms 1998), but in our study the lack of differences in pastures with and without grazing indicates that a moderate stocking rate of 0.6 AUM ha⁻¹ sustained over a 30-yr period did not reduce the stability of aggregates.

It is generally understood that formation of larger aggregates is enhanced by fine roots and fungal hyphae, while microaggregates are stabilized by long-chained organic compounds (e.g., polysaccharides) and fungal hyphae. Correlations with root biomass were significant and positive for the MWD and the 2–5-mm size fraction, and significant and negative for the <0.25-mm size fraction (Table 3). This supports the hypothesis that roots act as temporary binding agents which aid in stabilizing larger aggregates (Tisdall and Oades 1982). The correlations with polysaccharides and MWD and individual aggregate size fractions were all statistically significant, with the exception of the 0.25- to 1-mm size fraction (Table 3). Larger size fractions (2–5 mm and 1–2 mm) were negatively correlated and the smallest (<0.25 mm) fraction was positively correlated to polysaccharides. This is in agreement with findings by Six et al. (1998) and the general understanding that soil polysaccharides enhance formation and stabilization of soil microaggregates (Gregorich et al. 1994).

POTENTIAL MANAGEMENT IMPLICATIONS AND CONCLUSIONS

In British Columbia, fall grazing during the dormant season is often implemented in sensitive areas to increase the abundance of late seral grass species such as

bluebunch wheatgrass, the dominant grass species of the lower and middle elevation grasslands. Our results point to another benefit of long-term fall grazing done at a moderate stocking rate: lower soil compaction as indicated by lower bulk density and mechanical resistance. Interestingly, the elimination of grazing on these semi-arid grasslands for either 20 or 30 yr did not result in differences in soil total carbon and nitrogen or root biomass between spring and fall grazed pastures or between pastures with and without grazing. However, relative to early-season grazing, long-term fall grazing led to an increase of the labile pool of soil organic matter as reflected by greater soil polysaccharides. Our study showed that long-term grazing at a moderate stocking rate of 0.6 AUM ha⁻¹ did not have detrimental effects on soil properties as some land managers and ranchers have anticipated (Wikeem and Wikeem 2004). In fact, the exclusion of grazing for either 20 or 30 yr did not lead to any substantial improvements of soil properties relative to the pastures with grazing. In turn, it is very unlikely that long-term grazing at this stocking rate would lead to negative impacts on water quality (which is directly related to soil properties such as bulk density, mechanical resistance, aggregate stability, and organic matter content).

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